



Crystal growth principles, methods, properties of silicon carbide and its new process prepared from silicon cutting waste

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ABSTRACT

The third-generation semiconductor silicon carbide (SiC) has attracted widespread attention due to its excellent properties, such as high thermal conductivity, large bandgap, high breakdown field strength, and high saturation electronic drift rate, etc. Consequently, the growth process, physical structure, and properties of SiC crystals have also become research hotspots in industry and academia sectors. With the concept of carbon peak and carbon neutrality, the photovoltaic industry has witnessed rapid development. In the process of silicon wafer production, nearly half of the crystalline silicon is lost in the form of silicon powder into silicon cutting waste (SCW), which results in a great waste of resources and severe environmental pollution, and therefore the use of SCW for the preparation of SiC materials has received great attention in recent years. This paper highlights the principles and methods of SiC growth, crystal structure and properties, and discusses the application of SiC prepared from SCW.

1. Introduction

In recent years, silicon carbide (SiC) has evolved from a high-potential wide-band semiconductor to a widely recognized and indispensable material for power electronics [1]. Compared to the first generation of semiconductor materials represented by Si and Ge and the second generation of semiconductor materials represented by GaAs and InP, the third generation of semiconductor materials represented by SiC and GaN are endowed with the advantages of a wide bandgap, high thermal conductivity, high breakdown field strength, high saturation electronic drift rate and strong radiation resistance [2–4]. Therefore, devices prepared with the third-generation semiconductor materials could not only achieve stable operation at higher temperatures, but are also suitable for high-voltage and high-frequency scenario [5], as well as obtaining higher operational capabilities with lower power consumption.

SiC, which is also known as adamantite, has a density of $3.2 \text{ g}\cdot\text{cm}^{-3}$. Natural SiC is very rare and could be mainly obtained by artificial

synthesis [6]. SiC is an IV-IV compound composed of Si and C with a variety of isotopic types, SiC can be divided into two categories according to different crystal structures: α -SiC and β -SiC [7]. SiC powder is formed through process of the crystal growth [8], cutting [9], polishing [10] and other aspects of the formation of the substrate, in which the growth of SiC crystals is the core process, the core difficulty in improving the yield. The main crystal growth methods for SiC include physical vapor transport growth (PVT) [11], high temperature chemical vapor deposition growth (HTCVD) [12] and topseeded solution growth (TSSG) [13].

With the change of the world's energy structure, the exploitation of new renewable energy sources to replace fossil energy sources has become a hot research topic [14–16], and photo voltaic (PV) power generation is considered one of the most promising renewable and non-polluting energy technologies [17]. Therefore, the PV industry has grown rapidly over the past few decades [18], which has led to a rapid increase in the amount of silicon waste generated during the development of the PV industry. It has been reported that about 40% of high

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purity silicon are wasted in the form of scrap during the wafer production process [19], resulting in severe environmental damage and waste of resources. Therefore, the recycling of silicon cutting waste (SCW) can not only reduce the waste of resources, alleviate environmental pollution and improve the utilization rate of waste materials, but also generate considerable economic benefits. The recycling of SCW is in line with the principle of circular economy, creating a sustainable closed-loop system, and the use of SCW to prepare SiC materials is considered one of the effective ways to recycle SCW. At present, the main method of preparing SiC from SCW is the carbothermal reduction method, which has the advantages of low raw material cost, simple preparation process and fine SiC particles, but there are still problems such as high melting temperature, high energy consumption and difficulty in purification.

2. Crystal structure of SiC

SiC is a covalently bonded IV-IV compound, as shown in Fig. 1, the Si and C atoms in SiC are always in tetrahedral coordination, and each Si (or C) atom is connected to four C (or Si) atoms with sp^3 hybrid bonding, and the Si-C bonding energy is as high as 4.6 eV [20], which makes the structure of SiC extremely stable. The distance between the neighboring Si atoms or C atoms is about 3.08 Å. The Si (or C) atom is located at the center of the tetragonal structure outlined by the four neighboring C (or Si) atom, therefore, the distance between a Si atom and each C atom is the same, at 1.89 Å [21]. Since Si and C atoms are always tetrahedrally coordinated, the way in which the tetrahedra are linked and stacked is somewhat restricted. As shown in Fig. 2, there are only two possible ways to satisfy the tetrahedral coordination of SiC.

The close-packed structure between Si and C atoms is considered to be a periodic SiC crystal structure. So far, more than 250 different crystal systems have been discovered, with repetitions of the stacking sequence ranging from two to hundreds of layers [23,24]. The type of SiC can be labelled as nX , where, "X" denotes the Bravais lattice and "n" represents the number of diatomic layers contained in a lattice period [25]. The main polytypes of SiC are α -SiC (high-temperature stable hexagonal crystal system) and β -SiC (low-temperature stable cubic crystal system), β -SiC is stable below 2100°C, and starts to transform to α -SiC above 2100°C [26]. The most common crystal types in α -SiC are 2H-SiC, 4H-SiC, 6H-SiC, and 15R-SiC, and only one crystal type of β -SiC is 3C-SiC [7]. where, H, C, and R stand for hexagonal, cubic, and rhombohedral crystal systems, respectively. SiC polytypes differ in the stacking sequences of consecutive double layers of Si and C atoms, but only a few of them have been studied in detail, and four of these stacking sequences are considered to be the "basic" SiC polytypes which are the aforementioned 3C-SiC, 4H-SiC, 6H-SiC, and 15R-SiC [22]. From Fig. 3, it is clear that the stacking sequences of the cubic crystal system (3C) are ABC ...; the stacking sequences of the hexagonal crystal systems (4H, 6H) are ABCACB ... and ABCACB ...; The stacking sequences of the rhombic crystal system (15R) are ABCACBCABACBCB Fig. 4 shows the stacking sequences of the tetrahedral layers corresponding to the four "basic" SiC polytypes. The physical and chemical properties of SiC is strongly affected by its polytypes, which give it a wide range of thermal, electrical and mechanical properties [4].

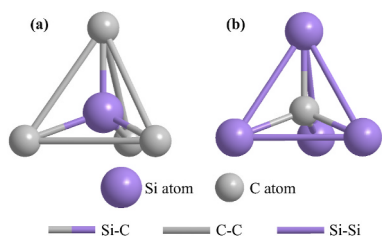


Fig. 1. Si and C atoms in simple tetrahedral coordination [22].

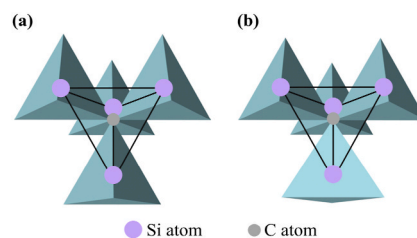


Fig. 2. Double tetrahedral coordination of Si (in purple) and C (in grey) atoms: (a) two tetrahedral layers with the same relative orientation; (b) two tetrahedral layers rotated by 180° relative to each other [22].

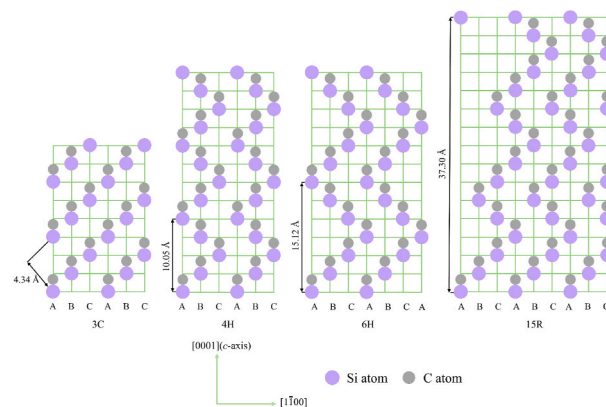


Fig. 3. Crystallographic stacking sequences of the 3C-, 4H-, 6H- and 15R-SiC double layers [7].

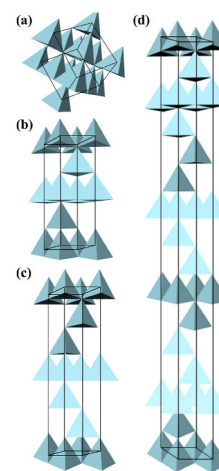


Fig. 4. Stacking sequences of tetrahedral layers corresponding to 3C-, 4H-, 6H- and 15R-SiC polytypes: (a) 3C, (b) 4H, (c) 6H and (d) 15R. Unit cells not drawn to scale.

The 3C polytype crystallizes in a cubic crystal system as shown in Fig. 5(a). It is the only sequence of two tetrahedral stacks with the same relative orientation as in Fig. 2. The lattice parameters are $a_0 = b_0 = c_0 = 4.359$ Å. β -SiC has a face-centered cubic lattice, but the asymmetric unit consists of one Si atom and one C atom, and the unit contains eight atoms (four Si and four C), so the density is $3.216 \text{ Mg}\cdot\text{m}^{-3}$ [22].

4H and 6H are hexagonal crystal systems with a wurtzite crystal structure, as shown in Fig. 5(b) and (c), with densities of $3.239 \text{ Mg}\cdot\text{m}^{-3}$ and $3.213 \text{ Mg}\cdot\text{m}^{-3}$, respectively. 15R polymorph has a rhombohedral crystal system, as shown in Fig. 5(d), with a density of $3.220 \text{ Mg}\cdot\text{m}^{-3}$. The lattice parameters of the α -SiC polymorphs are the same on the basal plane, i.e., $a_0 = b_0 = 3.073$ Å, but they are essentially different in the

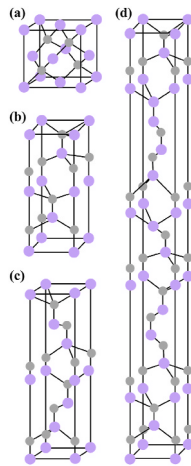


Fig. 5. Crystal structures of 3C-, 4H-, 6H- and 15R-SiC polytypes: (a) 3C, (b) 4H, (c) 6H and (d) 15R. Unit cells not drawn to scale.

lattice parameter c_0 [22].

3. SiC crystals growth principles and methods

As can be seen from the phase diagram of SiC (Fig. 6), there is no solid solution between Si and C [27], SiC crystals grow in the stoichiometric Si-C vapor and Si-C melt. As shown in the green curve in Fig. 6, the SiC source material sublimes in the temperature range of approximately 1800–2700°C under specific experimental conditions. Recrystallisation at a slightly lower temperature can produce a single crystal of SiC, which can be grown with the method of physical vapor transport growth (PVT) or high temperature chemical vapor deposition growth (HTCVD) (see Section 3.1 and 3.2). Another method is to use the solubility line of SiC in Si for solution growth. The SiC source material melts in the temperature range of about 1414–2870°C and recrystallizes at a slightly lower temperature to obtain SiC crystals, as shown by the red curve in Fig. 6, top-seeded solution growth (TSSG) can be used (see Section 3.3). Table 1 shows a comparison of the main typical parameters of SiC crystals growth.

3.1. Physical vapor transport growth

PVT is the most commonly adopted method for the growth of SiC crystals. It mainly includes three steps: sublimation of the SiC source, transport of the sublimates and surface reaction and crystallization of the seed crystal. The earliest systematic research on the growth of SiC crystals for electronic applications dates back to 1955, when Lely [34] first invented a vapor growth method and completed the growth of

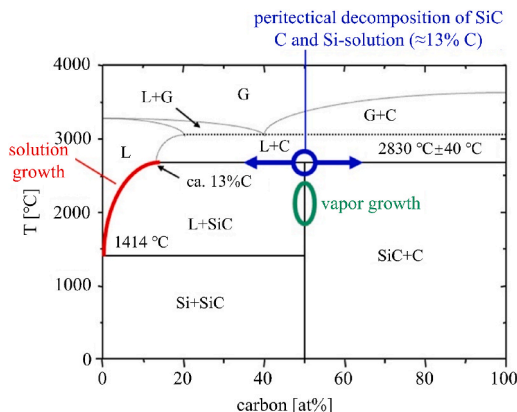


Fig. 6. Phase diagram of SiC [1].

Table 1

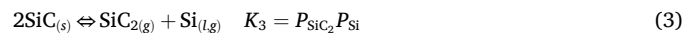
Typical parameters of different growth methods for SiC [28–33].

	PVT	HTCVD	TSSG
Size/mm	150 (6 inches)	100 (4 inches)	100 (4 inches)
Temperature/°C	2100–2400	2200–2500	1700–2200
Growth rate/mm·h ⁻¹	0.2–2.0	0.6–1.2	0.4–1.0

single crystal SiC, which is known as the Lely method and is a milestone in crystal growth. In 1978, Tairov et al. [11] developed the seed sublimation growth method, which is still widely used today and also known as the modified Lely method. Ziegler et al. [35] made a significant contribution to this new growth method, which is now known as the PVT. Although PVT related research has been conducted for a long time, the typical growth rate of SiC grown by PVT is about 0.3–0.5 mm/h, which is less than 1/100 of the growth rate of crystalline silicon. This is one of the biggest challenges that affect the development of PVT. Improving and stabilizing the crystal quality is another key issue remains to be solved for the growth process of SiC bodies [36].

Fig. 7 shows a schematic diagram of the reactor and gas phase transport used to grow SiC crystals for PVT, showing the temperature distribution and coil position. The source material used for SiC growth is usually high-purity polycrystalline SiC powder. After the reaction furnace is evacuated in a semi-closed graphite crucible, gases such as N₂, O₂, Ar, CO₂ and Ne are introduced, subsequently the high-purity SiC source material is sublimed by induction heating or resistance heating at a very low pressure (10–90 Pa) and a high temperature of about 2352°C [37]. The large temperature difference between the source material in the furnace and the seed crystal creates a gas density gradient, which leads to flow recirculation (Fick's diffusion and convection) [38].

The sublimation reaction can be expressed as follows:



The species of vapor in the PVT are mainly composed of Si atoms, Si₂C and SiC₂ molecules, and there is almost no molecular SiC in the gas phase [39]. The driving force for the growth of SiC is the partial pressure difference of the various gases (Si, Si₂C and SiC₂) between the hot SiC source and the slightly cooler SiC seed crystal [40]. The growth rate of SiC is therefore a function of the temperature of the source material, the distance from the source material to the seed crystal, the temperature gradient between the SiC source and the SiC seed, and the pressure of the inert gas applied. When the temperature gradient between the seed

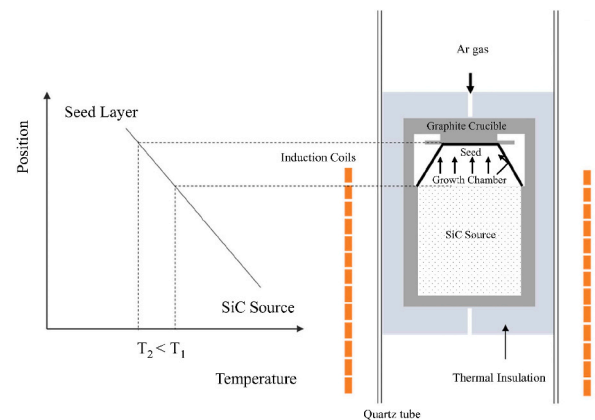


Fig. 7. Schematic illustration of the physical vapor transport (PVT) reaction furnace for SiC crystals growth, showing temperature distribution and coil positions [38].

crystal and the source material is constant, the growth rate tends to increase as the temperature of the seed crystal increases. In addition, when the temperature of the seed crystal is fixed, the growth rate increases with increasing temperature gradient, as shown in Fig. 8 [41].

Karpov et al. [43] calculated the partial pressures of species under C-rich SiC-C and Si-rich SiC-Si conditions as a function of temperature, as shown in Fig. 9. It can be seen that Si evaporates preferentially during the heating process, which causes the gas phase during sublimation to be generally Si-rich, making the source material that has not evaporated become much more C-rich, which in turn causes the source material to crystallize or graphitize during the growth process, as shown in Fig. 10 (a). Graphitization occurs at the edges and corners of the source powder and crucible, while crystallization occurs at the top of the powder (Fig. 10(b) shows a model of powder graphitization) [44]. As the degree of crystallization or graphitization increases, the growth rate of SiC crystals decreases over time. To prevent this from happening and to avoid carbonaceous inclusions in the grown crystals, the source material needs to be doped with Si to maintain a certain stoichiometric ratio or be Si-rich. Crystallization or graphitization of the source material can also be avoided to a certain extent by using TaC-coated crucibles [45]. If no significant crystallization or graphitization of the SiC source material is observed during sublimation and no incorporation of Si droplets or C inclusions is found during the crystallization of SiC, then both sublimation and crystallization take place with the same flux of Si and C [1].

The temperature distribution of the powder plays an important role in the stability of the sublimation of the powder and the growth rate of the crystals [46]. When the transport channels for the substances in the powder are unobstructed, a large temperature difference in the powder is beneficial for improving the sublimation efficiency and growth rate. At the same time, recrystallization at the bottom of the powder should also be avoided. When the sublimation of the powder is superior to the diffusion of the vapor in the gap between the powder and the growing crystals, it will hinder the transport channels for the substances in the powder and even interrupt the growth of the crystals [47]. When the growth temperature reaches 2200–2400°C, the argon pressure is reduced to 1–s 40 Torr, and the crystal begins to grow at a rate of 0.2–2 mm/h.

Simulation studies have proven to be key to the study of the thermodynamics and kinetics of the PVT growth of SiC [48–50], and have made important contributions to the study of the vapor species balance, crystallization, graphitization of the source powder, growth rate and temperature distribution [51]. Lu et al. [52] assumed that the SiC raw material was spherical particles with a diameter of 0.24 mm, and the diameter of the SiC particles did not change after being subjected to graphitization, and that the raw material area was a porous medium area. They considered the transport of gas phase components in the raw material as the flow and diffusion of gas in a porous medium composed of spherical particles, and derived a description of the transport of gas

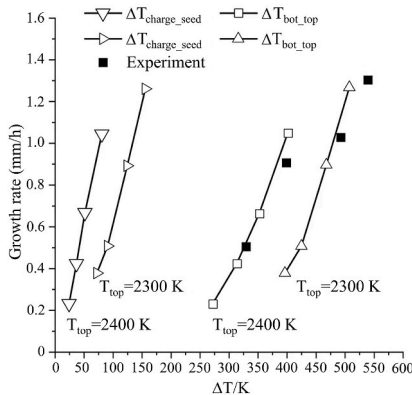


Fig. 8. Growth rate versus seed temperature and temperature gradient between seed and powder. An experimental data (after Ref. [42]) is also shown [41].

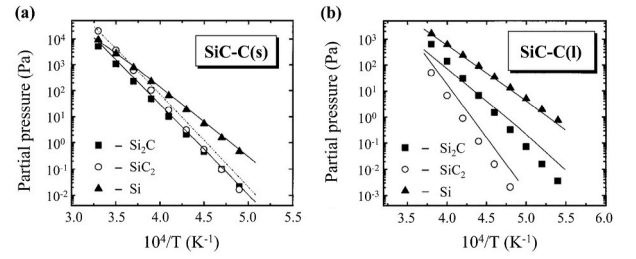


Fig. 9. Partial pressures of Si, Si₂C and SiC₂ as a function of temperature: (a) SiC-C; (b) SiC-Si [43].

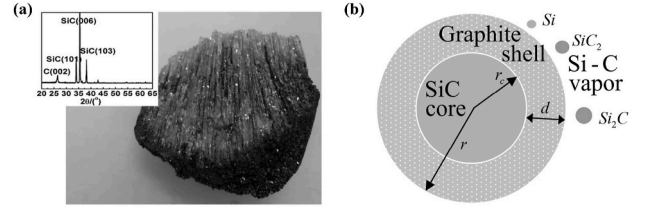


Fig. 10. (a) Crystallization and graphitization of SiC powder; (b) Modelling the graphitization of SiC particles [44].

phase components with the adoption of the modified continuity equation, momentum equation, and component equation:

$$\frac{\partial \rho_g}{\partial t} + \nabla \cdot (\epsilon \rho_g \vec{u}) = \epsilon \sum \dot{R}_i \quad (4)$$

$$\frac{\partial \rho_g \vec{u}}{\partial t} + \nabla \cdot (\epsilon \rho_g \vec{u} \vec{u}) = -\epsilon \nabla P + \nabla \cdot (\epsilon \tau) - \epsilon \rho_g \vec{g} - \frac{\epsilon \mu}{K} \vec{u} \quad (5)$$

$$\frac{\partial \epsilon \rho_g Y_i}{\partial t} + \nabla \cdot (\epsilon \rho_g \vec{u} Y_i) = \nabla \cdot (\epsilon \rho_g D_{\text{eff}} \nabla Y_i) + \epsilon \dot{R}_i \quad (6)$$

Where, ρ_g denotes the average density of the gas phase; μ is the viscosity coefficient; ϵ represents the dielectric constant; $\dot{R}_i$ is the reaction rate of component i ; K denotes the Darcy law permeability coefficient; D_{eff} is the effective diffusion coefficient. Calculated separately by the following formula:

$$K = \frac{(2r)^2 \epsilon^3}{175(1 - \epsilon)^2} \quad (7)$$

$$D_{\text{eff}} = \frac{1}{\tau} D_{ij} = \sqrt{\epsilon} D_{ij} \quad (8)$$

Where, τ denotes the tortuosity of the porous medium [53], which is related to the shape of the porous medium matrix and the porosity; D_{ij} is the binary diffusion coefficient of the gas.

Stefan flow exists at the crystal growth interface, and the component boundary conditions use the Hertz-Knudsen model, which has the following components for SiC vapor and Ar:

$$J_i = v_n c_i - D_{ij} \frac{\partial c_i}{\partial n} = \frac{P_i - P_i^*}{\sqrt{2\pi M_i R T}} \quad (9)$$

$$J_{\text{Ar}} = 0 \quad (10)$$

Where, J is the molar flow; c_i is the molar concentration of i component; v_n is the normal velocity; P_i is the partial pressure of component i on the particle surface; and P_i^* is the equilibrium partial pressure ($P_i^* = \exp(-\Delta G/RT)$). Neglecting the radial growth of the crystal, the normal growth rate G of the crystal is:

$$G = J_i \frac{M_{\text{crystal}}}{\rho_{\text{crystal}}} \quad (11)$$

Where, M_{crystal} is the molar mass of SiC crystals and ρ_{crystal} is the density of SiC crystals. With the adoption of the moving grid technique, the node positions of the crystal growth interface are adjusted according to the values calculated in the previous step before each time step, simulating the changes in the shape of the crystal growth interface and the thickening of the crystal, and reconstructing the body grid on both sides of the crystal growth interface.

3.2. High temperature chemical vapor deposition growth

The concept of HTCVD was proposed by Kordina and others in the mid-1990s [12]. In terms of the growth principle, HTCVD and PVT have the same mechanism, in which the vapor is the intermediate source for SiC deposition. The difference is that the initial source materials for the PVT and HTCVD are SiC powder and gas, respectively. Therefore, the HTCVD is also known as the gas source method for the growth of SiC. One advantage of the HTCVD over the PVT is that the Si/C ratio in the vapor can be more easily controlled by adjusting the ratio and flow rate of the source gas, which effectively suppresses the deterioration of crystal quality caused by the Si/C ratio, thereby reducing the cost of SiC production, improving the crystal quality of the wafer and increasing the growth rate of SiC [51]. Hoshino et al. [54] found that the growth rate increased linearly with increasing partial pressures of the source gases. At an input SiH_4 pressure of 10.1 kPa, the growth rate reached a high speed of 3.1 mm/h, which is about 2–10 times the rate of PVT growth of SiC. Therefore, it is a promising technology for the rapid growth of SiC and is expected to replace the widely used PVT for growing SiC. The main reason accounting for the high growth rate is the unique cluster structure, vapor transport mechanism and continuous gas supply system, however, HTCVD still remains underdeveloped [55]. At the same time, the density of residual impurities such as N and B in SiC produced by the HTCVD is around 10^{14} cm^{-3} , which makes it easier to produce semi-insulating wafers [56,57].

The schematic diagram of the HTCVD reactor is shown in Fig. 11. The source gas is supplied from the bottom of the reactor and decomposed at high temperatures (2500°C) by an induction coil. The decomposed gas then comes into contact with the seed crystals, which are slightly below the gas temperature, and subsequently crystallizes [58]. The reactor designed by Tsuchida et al. [36] is shown in Fig. 12. By controlling the conditions in the reactor, such as the size of the central plate and the rotation speed of the seed crystal (diffusion control), the growth rate of high-quality SiC can be controlled to achieve a high growth rate of about 3 mm/h or higher value. Hoshino et al. [59] grew high-quality crystals in the crystal plane zone at a seed temperature of 2350°C and an ultra-conventional growth rate of 5.4 mm/h without macro-steps or voids. In HTCVD, conditions such as temperature distribution, fluid flow profiles, chemical reactions, and partial pressures of gases need to be taken into consideration. At present, the systems to grow SiC with the method of HTCVD that have attracted the most interests of researchers are usually ternary (e.g. Si-C-H) or quaternary (e.g.

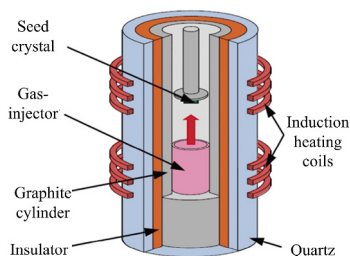


Fig. 11. Schematic diagram of the high temperature chemical vapor deposition reactor [58].

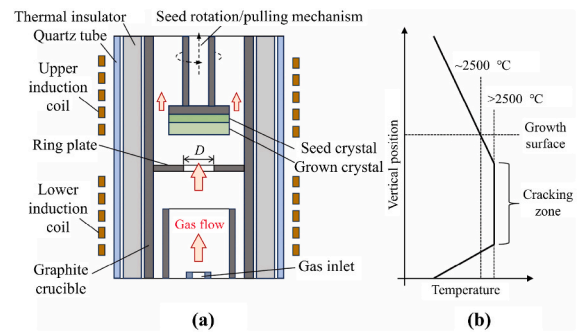


Fig. 12. Schematic diagram of the high temperature chemical vapor deposition reactor designed by Tsuchida et al. [36].

Si-C-H-C) systems. Common Si-C-H systems include the $\text{SiH}_4\text{-C}_3\text{H}_8\text{-H}_2$ system [58,60–62] and the TMS-H_2 one [63–66].

During the growth process of SiC on HTCVD, silane gas is directly pyrolyzed to form Si clusters, which then react with hydrocarbons (C_xH_y) in carbon-bearing precursor gases (e.g. propane) to form SiC bulk crystals [55]. The direct pyrolysis of the carrier gas makes HTCVD more energy efficient than PVT. At the same time, the high-speed carrier gas also helps these clusters reach the substrate, so HTCVD grows SiC faster than other methods [55,67]. Therefore, the growth rate depends largely on the number of SiC clusters present on the growing SiC substrate.

In the $\text{SiH}_4\text{-C}_3\text{H}_8\text{-H}_2$ system, for example, the reaction gas SiH_4 is carried by H_2 or He , mixed with C_3H_8 on the way, and then introduced into the reactor together. The reaction gas decomposes at high temperatures to form SiC, which adheres to the surface of the substrate material and continues to grow along the surface of the material. The residual gas produced in the reaction is treated in a waste treatment device and discharged [68]. Danielsson et al. [69] and Kito et al. [60] calculated the partial pressures of the main substances from the non-limited reaction rate of the gas-phase reaction, which is expressed in the form of the Arrhenius equation (Eq. (12)) [70], where A is the pre-exponential factor. Fig. 13 shows a comparison of the partial pressures of the main species calculated by different researchers. The main gas phases identified were SiC_2 , Si_2C , Si, and C_2H_2 , which are consistent with the thermodynamic equilibrium calculations of Stinspring et al. [71]:

$$K = AT^n \exp(-E_n / RT) \quad (12)$$

The growth rate of HTCVD can be expressed by the following formula [61]:

$$Gr \propto [(p_x - p_e) / (RT)] [D / \delta] \quad (13)$$

Where, p_x is the partial pressure of the source gas; p_e denotes the equilibrium vapor pressure; R is the gas constant; T represents the temperature; D is the diffusion coefficient in the gas boundary layer; and δ is the gas boundary layer. Partial pressures p_x depend on the gas flow ratio, as shown in the table below:

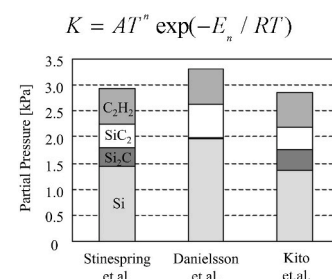


Fig. 13. Partial pressures of the main substances calculated by equilibrium reaction modelling and gas phase reaction modelling at 2500°C [60,69,71].

$$p_x \propto [(f_{\text{SiH}_4} + f_{\text{C}_3\text{H}_8}) / (f_{\text{SiH}_4} + f_{\text{C}_3\text{H}_8} + f_{\text{H}_2})] \quad (14)$$

Where, f_{gas} is the flow rate of the source gas. The gas flow ratio is defined as: $(f_{\text{SiH}_4} + f_{\text{C}_3\text{H}_8}) / (f_{\text{SiH}_4} + f_{\text{C}_3\text{H}_8} + f_{\text{H}_2})$. The shape of the attached layer is expressed by the following equation:

$$(1/\delta) \propto (U_0 P)^{1/2} \quad (15)$$

$$(1/\delta) \propto (\omega P)^{1/2} \quad (16)$$

Where, P is the total pressure; U_0 is the gas flow rate; ω is the seed rotation speed. In order to achieve a high growth rate, it is necessary to increase $(p_x \cdot p_e)$ or (D/δ) to equation (13).

As can be seen from Fig. 14(a), the growth rate tends to increase with the increase of the flow rate ratio, that is, the growth rate under isothermal conditions is proportional to p_x , which is consistent with the linear relationship of Eq. (13). Increasing the partial pressure can effectively increase the growth rate. In addition, the growth rate tends to decrease with increasing temperature, which means that the rate of sublimation etching from the surface increases with increasing surface temperature. At the same time, it can be seen from Eq. (13) that, in order to increase the growth rate, the area of the boundary layer needs to be reduced. From Eq. (15) and Eq. (16), it can be seen that under constant pressure, the centrifugal equation is a function of the gas flow rate U_0 or ω the seed rotation speed. U_0 or ω needs to be increased so as to reduce the centrifugal equation. Fig. 14(b) shows that the growth rate increases proportionally with $1/\delta$, which is corresponding to the linear relationship in Eq. (13). The maximum growth rate reaches 3.6 mm/h, which is significantly higher than that of the PVT.

Fig. 15(a) shows the 4H-SiC crystals with the thickness of 17 mm which were grown at a growth rate of 1.9 mm/h, in which no other polycrystalline inclusions or cracks were found. Fig. 15(b) and (c) show the dislocation density of the wafers evaluated by KOH molten etching. The wafers with a dislocation density of 1700 and 11550 cm^{-2} were adopted as seeds, respectively. The quality of crystal obtained by growing at the speed of 1.0 mm/h was very high, with an evaluated dislocation density of 1060 cm^{-2} , while the dislocation density of the crystal grown at a growth rate of 3.0 mm/h was evaluated at 3020 cm^{-2} . It can be seen that the dislocation density of the grown crystals is lower than that of the seed crystals. In particular, there exists a significant difference between the dislocation density of the crystals grown at a growth rate of 3.0 mm/h and that of the seed crystals, which indicates that 4H-SiC crystals can be grown with a reduced or not increased dislocation density even at a high growth rate of 3.0 mm/h [62]. This result shows that HTCVD is a promising method for the growth of SiC. High growth rates can improve the production efficiency of SiC bulk crystals and help reduce the energy consumption in the production process of SiC crystals. The quality of the crystal can be improved by reducing the dislocation density during the growth stage or making the dislocation density of the grown crystal lower than that of the seed crystal. However, expanding the diameter and length of the crystal still remains a key challenge for HTCVD to catch up with PVT.

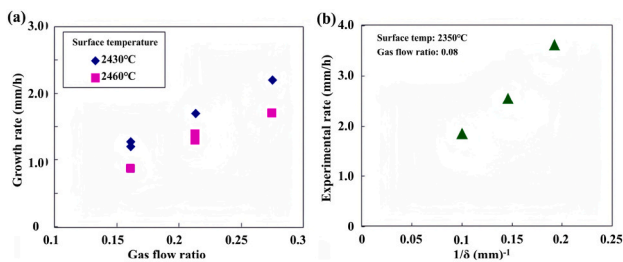


Fig. 14. (a) The relation of the growth rate and the gas flow ratio as a function of the surface temperature; (b) The relation between the experimental growth rate and the reciprocal boundary layer $1/\delta$ [61].

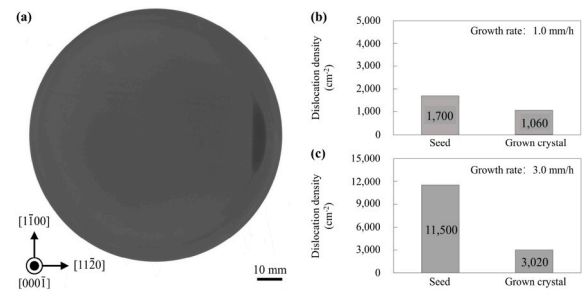


Fig. 15. (a) 4H-SiC crystal with a thickness of 17 mm grown at 1.9 mm/h; (b) KOH molten etching evaluation of crystal grown at 1.0 mm/h; (c) KOH molten etching evaluation of crystal grown at 3.0 mm/h [62].

3.3. Top seeded solution growth

It has been reported that TSSG can produce SiC crystals with higher quality [33,72,73]. Kusunoki et al. [74] demonstrated that basal plane dislocations (BPDs) present in the seed crystal do not propagate into the growing crystal and no additional BPDs are formed during the solution growth process. In addition, apart from inheriting intrinsic dislocation defects from the seed, there are almost no new threading screw dislocations and threading edge dislocations (TDs) generated [75]. Compared with the traditional PVT method, the reason for the excellent quality of the low defects SiC crystals grown by TSSG lies in the fact that the growth temperature is lower, the temperature gradient at the growth interface is smaller, and the growth is carried out under the conditions close to thermodynamic equilibrium. The low growth temperature and conditions close to thermodynamic equilibrium enable defects in the crystal to spread laterally, thus significantly improving the crystal quality [76]. In addition, microtubules, known as "killer defects", can be completely eliminated during the solution growth process, which results in a microtubule-free SiC body crystal [77]. TSSG grown SiC is also endowed with the advantages of strong controllability of the growth process, easy expansion, and easy p-type doping [78]. Based on these unique advantages of the TSSG method, researches on this method have been increased in recent years. TSSG growth of SiC has become one of the most competitive low-cost, high-quality SiC substrate innovation technologies. Fig. 16 shows the large-size SiC, as well as the diameter and thickness of TSSG grown in recent years.

Fig. 17(a) shows a schematic diagram of the TSSG SiC growth system. The main structure consists of an external induction heating coil, a graphite insulation blanket, a high-purity graphite crucible that also serves as a high-temperature solution container and C source, a high-temperature melt, SiC seed crystals and a suspension rod [82]. When growing SiC crystals, the crucible is first heated to a temperature at which the raw materials are completely melted using an induction coil.

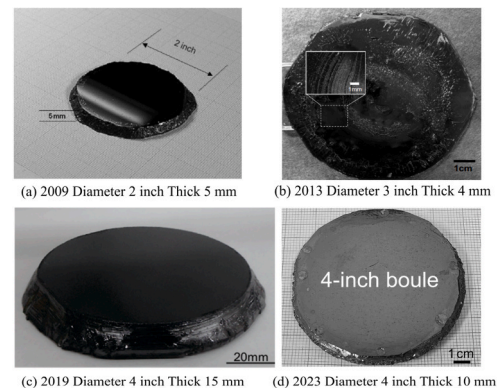


Fig. 16. Typical SiC crystals obtained by TSSG method in recent years and their sizes and thicknesses [79–82].

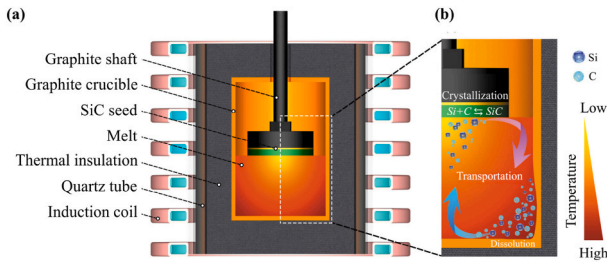


Fig. 17. Schematic diagram of SiC growth by TSSG. (a) Schematic diagram of the apparatus for SiC growth by TSSG; (b) Schematic diagram of the three basic growth processes of TSSG [82].

The temperature in the furnace is usually controlled in the range of 1700°C and 2100°C. After the formation of a stable temperature field, the concentration of C in the high-temperature solution reaches equilibrium, and the suspension rod with the seed crystal is slowly immersed in the high-temperature solution from the top [83,84]. During the growth process of SiC, the temperature field is controlled to produce a temperature gradient as shown in Fig. 17(b). Due to the temperature gradient, the solubility of C at the top and bottom of the crucible is different (the temperature near the seed crystal is lower, the solubility of C is lower, the temperature at the bottom is higher, and the solubility of C is higher). The melt at the bottom of the crucible first becomes saturated with C and is transported with the temperature gradient to the vicinity of the seed crystals, where it combines with Si in the high-temperature solution to grow SiC single crystals on the surface of the seed crystals [78]. When the C-saturated melt grows SiC, it becomes unsaturated and returns to the bottom of the high-temperature crucible to melt C again to form a saturated solution. The entire process repeats itself in a cycle. To achieve the rapid and continuous growth of SiC crystals, a continuous supply of carbon is required. The adoption of a high-purity graphite crucible as a carbon source is a simple and effective way to continuously supply carbon during the crystal growth process [85].

Compared to other methods, the growth rate of TSSG still remains relatively low. The establishment of appropriate temperature and flow field in the solution is the key to improve the crystal growth rate, and the appropriate temperature and flow field can promote the migration of carbon from the graphite crucible wall to the SiC growth interface, thus enhancing the concentration gradient of carbon at the growth interface and effectively improve the growth rate of SiC crystals [86,87]. When adopting the TSSG method, during the growth of SiC single crystals, the heat transfer and flow phenomena in the melt are controlled by the continuity equation, the momentum conservation equation, and the energy conservation equation [88]. The governing equations in the fluid are as follows [89]:

Continuity equation:

$$\nabla \cdot (\rho \mathbf{v}) = 0 \quad (17)$$

Momentum equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mu \nabla^2 \mathbf{v} - \beta_T (T - T_0) \mathbf{g} \quad (18)$$

Energy equation:

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \alpha \nabla^2 T \quad (19)$$

Where, ρ is the melt density; $\mathbf{v}$ is the melt velocity; t is the time; p is the pressure; μ is the melt dynamic viscosity; β_T is the melt thermal expansion coefficient; T is the melt temperature; T_0 is the lowest temperature in the melt; $\mathbf{g}$ is the acceleration of gravity; and α is the thermal diffusivity. The above equation is based on the following assumptions [90]:

- 1) Growing melt is a non-compressible Newtonian fluid.
- 2) Low solubility of C, without considering solute convection.
- 3) The temperature difference within the melt is sufficiently small to apply the Boussinesq approximation.
- 4) The deformation of the dynamic interface (due to convection in the melt) is negligible.

In the actual numerical calculations, the above equation may be appropriately adjusted according to the actual model, but the principle is basically similar. Table 2 lists the parameters of the high-temperature solution used in the calculation.

A sufficient concentration of C is a prerequisite for the rapid growth of SiC, but the solubility of C in pure Si solution is very low (usually less than 0.1% in the experimental temperature range) [91]. Therefore, adding one or more transition metals or rare earth elements (Cr, Ti, Y, etc.) as co-solvents to the Si melt can not only effectively reduce the growth temperature of the system, but also significantly increase the solubility of C in high-temperature solutions [92–94]. After decades of development, nearly 30 elements have been developed and successfully used in the growth of SiC crystals by TSSG, including but not limited to: Si-Li, Si-Ti, Si-Cr, Si-Fe, Si-Se, Si-Ni and Si-Co and other binary systems, Si-Gr-Al, Si-Cr-Ce, Si-Cr-Co, Si-Cr-Fe and other ternary systems, and Si-Cr-Al-Sn, Si-Cr-Al-Ce quaternary systems [95]. Fig. 18 shows that nearly 30 solution systems have been developed for the growth of SiC crystals.

Zhou et al. [97] studied the influence of co-solvent Cr and additive Al on cellular structures and inclusions formation through numerical simulation of phase field models. The experimental results show that the growth rate of the 40% Cr solution is three times higher than that of pure Si solution. The $\text{Si}_{0.59}\text{-Cr}_{0.4}\text{-Al}_{0.01}$ solution was shown to be a promising candidate for maintaining high growth rates while preserving step stability and effectively inhibiting cellular structures and inclusions formation. Kawanishi et al. [93] studied the availability of Cr-rich Cr-Si solvents for rapid solution growth of 4H-SiC using three different compositions of Cr-Si solvents. The solubility of C in the quasi-binary (Cr-37.5 mol% Si)-SiC system increased from 6.9 mol% at 1923 K to 11.8 mol% at 2273 K at 1933 K, the growth rates of SiC were 1220 and 1310 $\mu\text{m}\cdot\text{h}^{-1}$ when adopting Cr-27.5 mol% Si and Cr-45 mol% Si solvents, respectively. This indicates that Cr-rich Cr-Si solvents are more likely to achieve rapid growth of 4H-SiC than traditional Si-rich Cr-Si solvents. Li et al. [98] used the high-temperature chemical equilibrium method to determine the temperature dependence of the solubility of C in Si-Nd solutions and concluded that the composition of Si-Nd solutions suitable for the growth of SiC crystals is between 25 and 42 mol% Nd at the temperature of 1823 and 1923 K. Si-Nd solutions can not only increase the growth rate of SiC single crystals, but also improve their quality. While the Si-Nd solutions have large values of carbon supersaturation ΔC , they have similar values of σ , degree of supersaturation, compared to those of other conventional Si-based growth solutions. Harada et al. [99] comprehensively investigated the effects of Si-X components ($X = \text{Cr, Ti, Fe, Zr, Co, Mo, W, Y}$ and Ni) on carbon solubility and unintentional doping, as shown in Fig. 19(a) and (b). The results show that the Si-Y and Si-Zr systems exhibit higher carbon solubility and lower unintentional doping behavior than other systems.

Table 2

Physical properties of high temperature solutions used in the numerical simulation of the TSSG method [90].

Material	Property (Symbol)	Value	Unit
Si melt	Density (ρ)	2550	$\text{kg}\cdot\text{m}^{-3}$
	Viscosity (μ)	8.0×10^{-4}	Pa·s
	Electrical conductivity (σ_e)	1.2×10^6	$\text{S}\cdot\text{m}^{-1}$
	Thermal conductivity (κ)	65	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
	Specific heat (C_p)	1.0×10^3	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$
	Surface tension coefficient (σ_T)	-2.5×10^{-4}	$\text{N}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
	Emissivity (β)	1.4×10^{-4}	K^{-1}

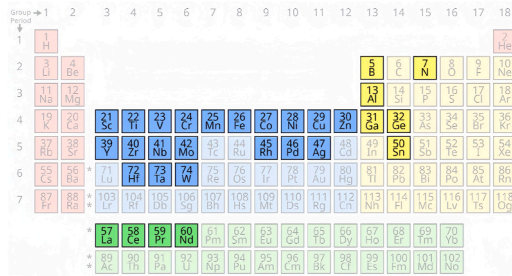


Fig. 18. Elements of silicon-based solvents applied to the solution growth of 4H-SiC single crystals [96].

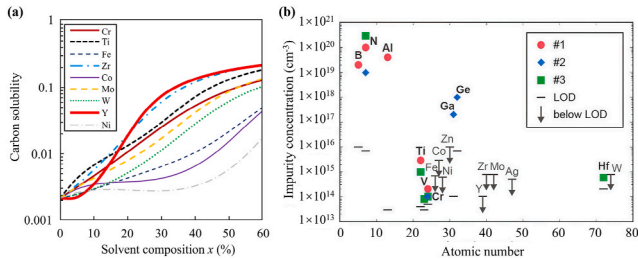


Fig. 19. (a) Carbon solubility of Si-X fractions (X = Cr, Ti, Fe, Zr, Co, Mo, W, Y and Ni) at 1800°C; (b) Unintentional doping behavior within the crystals [99].

However, considering that the Si-Zr system has a Zr-C stable phase in the high-temperature solution, which is not conducive to the continuous growth of crystals, the Si-Y system exhibits better overall performance. As research has progressed, the problem of carbon dissolution has been better resolved and is no longer a key factor that limits the maximum growth rate of crystals [100]. However, due to the high growth temperature and the difficulty of testing, the critical thermodynamic parameters and influencing factors for crystal growth, such as the freezing point, surface tension, viscosity and related phase diagram of the high-temperature solution in the TSSG for growing SiC single crystals, still remains unclear.

In this section, SiC growth techniques are mainly classified into three types, including PVT, HTCVD, and TSSG, in order to evaluate the advantages and disadvantages of the three methods, Table 3 lists the advantages and disadvantages of the three SiC crystal growth methods so as to compare their feasibility.

4. SiC materials properties

SiC exhibits unique and novel physical and chemical properties owing to its different polytypes. The properties of SiC materials depend on their strong covalent bonds and stable crystal structure [26]. SiC has high thermal conductivity, a large bandgap, a high saturation electronic drift rate, and a high breakdown field strength. It is widely used in high-temperature and high-frequency fields. This section mainly introduces the thermal properties (melting point, thermal conductivity, etc.), electrical properties (bandgap, electron mobility, breakdown field strength etc.), mechanical properties of SiC, as well as the effects of impurity doping on properties. Table 3 compares the properties of several important semiconductor materials.

4.1. Thermal properties

SiC has an extremely high melting point (2700°C) [101]. When heated in air, SiC could form a hard SiO₂ oxide film that prevents it from oxidizing for a short time (Hours) at 1850°C and for a long time (Days) at 1500°C [2]. At the same time, since the higher thermal conductivity of SiC than other semiconductors enable it applied for

Table 3

SiC growth methods and their pros, cons.

Growth Methods	Pros	Cons
PVT [34, 36,44]	The systematic research has an early origin, and the related research is more abundant. The source material is only SiC powder, which is simple in raw material composition and easy to control the growth process	Slower growth rates. Prone to graphitizing during growth leading to carbon inclusions in the grown crystals
HTCVD [51,62, 68]	Easier control of the Si/C ratio of the gas phase can effectively suppress the deterioration of crystal quality due to the Si/C ratio. It is about 2–10 times higher than the rate of SiC growth by PVT. Reduces the dislocation density in the growth phase so that the dislocation density of the grown crystals is lower than that of the seed crystals	Since the growth gas source material is a gas, tail gas treatment is required. Small size and short length of the grown crystals
TSSG [74–78, 91]	Reduction of dislocation and defect generation to improve crystal quality. The growth process is highly tunable, easy to expand, and easy to achieve p-type doping. The temperature during the growth process is much lower than that of the fumed phase method. Lower generation cost due to high purity silicon and high purity graphite crucible as raw materials	The lower solubility of carbon in the silicon melt leads to a slower growth rate. Due to the late development of TSSG, the key thermodynamic parameters and factors affecting the growth of SiC crystals by TSSG are still unclear

preparing high-temperature devices. However, the high local heat flow causes the device to overheat, and the increase in device temperature reduces their performance and reliability, making thermal management a huge challenge [102,103]. High thermal conductivity (κ) is essential in the thermal management design of these electronic and optoelectronic devices, especially for high-power devices [104,105].

The high thermal conductivity of SiC is resulted from the significant contribution of phonons. The stiff bonding, light constituent atoms and relatively low electron concentrations result in a lattice vibration (phonon) contribution to thermal conductivity that is much greater than that of the charge carriers [106]. The thermal conductivity can be expressed in terms of the effective phonon relaxation time ϑ_{pr} and is given as follow [107,108]:

$$k(T) = (4\pi k_B / \nu_s) (k_B T / h)^3 \times \int_0^{\theta/T} x^4 \vartheta_{pr}(x) [e^x / (e^x - 1)] dx \quad (20)$$

Where, θ is the Debye temperature; ν_s is the average velocity of phonons; h is Planck's constant; k_B is Boltzmann's constant, and $x = (\hbar\omega)/(2\pi k_B T)$. The phonon time ϑ_{pr} in the above equation can roughly be obtained based on the relaxation time approximation (RTA). In the RTA, each phonon scattering mechanism is considered to make a separate and independent contribution to the overall relaxation rate ϑ_{pr}^{-1} . Thus, for the i th scattering mechanism, the following equation applies:

$$\vartheta_{pr}^{-1} = \sum_i \vartheta_i^{-1} = \vartheta_I^{-1} + \vartheta_U^{-1} + \vartheta_D^{-1} + \vartheta_B^{-1} + \vartheta_N^{-1} + \vartheta_{4P}^{-1} \quad (21)$$

Where, ϑ_I^{-1} , ϑ_U^{-1} , ϑ_D^{-1} , ϑ_B^{-1} , ϑ_N^{-1} , ϑ_{4P}^{-1} are the relaxation times associated with isotope scattering, the Umklapp process, dislocation scattering, the phonon-boundary interactions, normal phonon processes, and four-phonon events, respectively.

Joshi et al. [109] investigated the temperature dependence of the

thermal conductivity of SiC under different phonon scattering mechanisms based on the above formula, as shown in Fig. 20. The four-phonon process is necessary to obtain thermal conductivity values that match experimental results well. In addition, the scattering of phonons by dislocations resulting from the internal lattice strain was also studied. The calculation results show that dislocations may adversely affect thermal conductivity at lower temperatures. The study by Joshi et al. contributes to the understanding of phonon scattering mechanisms in thermal conductivity calculations. However, it is limited by the incorporation of numerous empirical parameters, which restricts the accuracy of thermal conductivity predictions.

4.2. Electrical properties

As a third-generation semiconductor, SiC exhibits excellent electrical properties and is widely used in high-frequency semiconductor devices [111–113]. The lattice parameters of SiC are much smaller than those of Si and GaAs, which is the physical reason for the larger electron band gap E_G and breakdown field E_B . The larger E_G and smaller lattice parameter result in a larger effective electron mass, which in turn leads to a lower electron mobility μ_n in SiC. Due to the large E_B and high electron saturation speed v_s , high-frequency operation is not restricted, resulting in a high switching frequency of the power device [1]. The high breakdown field strength of wide band gap (WBG) semiconductors can reduce the thickness of the drift region and increase the doping concentration at a given rated voltage, thereby reducing the on-resistance of the power device. WBG with the same on-resistance can have a smaller area, which means a smaller capacitance. Smaller capacitance and higher saturation drift velocity allow higher switching speed and lower losses per switching cycle [114]. Fig. 21 shows the characteristics of WBG material.

B. J. Baliga derived the equation for the ideal specific on resistance based on the theoretical relationship between the specific on resistance of the drift layer and the basic properties of the material [116,117]:

$$R_{on,sp} = \frac{W_D}{q\mu_n N_D} = \frac{4BV^2}{\varepsilon_s \mu_n E_C^3} \quad (22)$$

Where, W_D is the thickness of the drift region for the required breakdown voltage (BV); q is the electron charge; μ_n is the electron mobility; N_D is the doping concentration of the drift region for the required BV; ε_s is the dielectric constant of the semiconductor, E_C is the critical electric field strength for the required breakdown voltage. In deriving this equation, the ideal parallel plane breakdown condition is assumed. Due to the development of edge terminal technology, the ideal breakdown voltage is close to the actual device [118].

The denominator of the above equation is often referred to as Baliga's figure-of-merit (BFOM) for power devices:

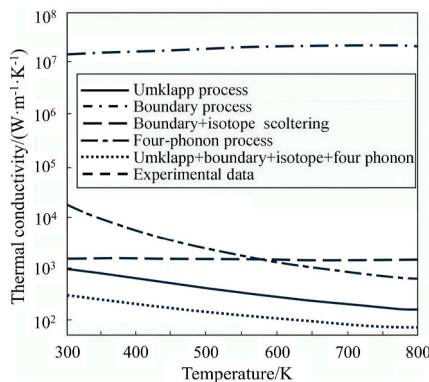


Fig. 20. Simulation results of SiC thermal conductivity versus temperature for different phonon scattering mechanisms. An experimental curve (after Ref. [110]) is also shown [109].

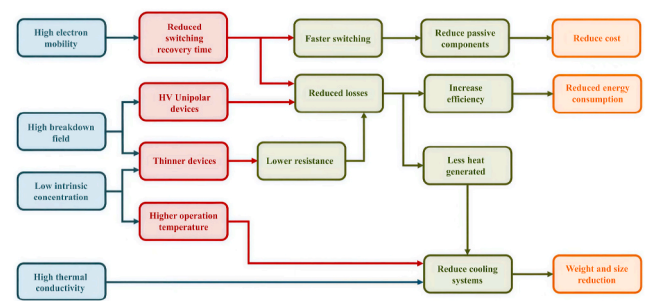


Fig. 21. WBG semiconductor parameter capabilities (blue); physical properties affected by WBG advantages (red); power electronics characteristics (green); and product benefits (yellow) [115].

$$BFOM = \varepsilon_s \mu_n E_C^3 = \frac{4BV^2}{R_{on,sp}} \quad (23)$$

The breakdown field strength of each semiconductor material varies with the BV [119]. Therefore, BFOM is applied to the devices made of different semiconductors with the same BV. From the above formula, it can be seen that BFOM can be calculated from BV and the on-resistance, in $W \cdot cm^{-2}$. This has been used to assess how close the actual device is to the ideal value [120]. Lower on-resistance results in a smaller forward voltage drop and reduces the thermal load on the system, improving the overall efficiency of the WBG semiconductor power device. In addition to saving energy, low-loss power devices could also help protect the environment since they minimize the use of fossil fuels to meet energy needs [121].

4.3. Mechanical properties

SiC is one of the hardest materials with a pressure of about 25 GPa, which is similar to B₄C [21]. Only diamond (60–120 GPa) and cubic boron nitride (borazon 40 GPa) are harder than SiC [122]. Table 4 shows the main mechanical properties of SiC and Si, which are not related to the polytypes of SiC. SiC can still maintain high hardness and elasticity at high temperatures [20]. The hardness (Mohs hardness), Young's modulus and yield strength of SiC are all higher than those of Si, making it more suitable for use in some harsh environments. The Poisson's ratio of SiC is similar to that of Si, making it similar in deformation when subjected to force (see Table 5).

4.4. Impurity doping effects on performances

Doping is regarded as a critical technology for introducing dopants to modify and improve the properties of SiC [127,128]. The diffusion of impurities in SiC usually requires high temperatures. The annealing temperature after implantation usually needs to exceed 1600°C to activate the implanted material and restore the crystal lattice. At the same time, effective sealing technology is used during the

Table 4
Comparison of the properties of several semiconductor materials [101,123].

Parameters	Si	GaAs	3C-SiC	4H-SiC	6H-SiC
Thermal conductivity (W·cm ⁻¹ ·K ⁻¹)	1.5	0.5	3.2	3.7	4.9
Bandgap (T < 5K)	1.12	1.43	2.40	3.26	3.02
Electron mobility (cm ² ·V ⁻¹ ·s ⁻¹)	1400	8500	800	1000	400
Hole mobility (cm ² ·V ⁻¹ ·s ⁻¹)	471	400	40	115	101
Saturation electronic drift rate (10 ⁷ cm·s ⁻¹)	1.0	2.0	2.5	2.0	2.0
Breakdown field strength (MV·cm ⁻¹)	0.25	0.3	2.12	2.2	2.5
Dielectric constant (F·m ⁻¹)	11.8	12.8	9.7	9.7	9.7

Table 5
Main mechanical properties of SiC and Si at room temperature [124,125].

Performances	4H-SiC or 6H-SiC	Si
Mohs hardness	9.2	7
Young's modulus (GPa)	390–690	160
Yield strength (GPa)	21	7
Poisson's ratio	0.21	0.22
c_{11}	501	166
c_{12}	111	64
c_{13}	52	–
c_{33}	553	–
c_{44}	163	80

(c_{ij} represents the elastic constant [126])

high-temperature implantation annealing process to maintain surface properties [129]. In the case of a small amount of impurities, the matrix usually does not lose its structural properties. There are various doping strategies according to the performance requirements of SiC. The types of doping include chemical doping, conductivity doping, magnetic doping, neutron transmutation doping, modulation doping, and doping in organic molecular semiconductors, etc [130]. To achieve the desired properties of SiC, the doping and impurity levels of the n-type, p-type and semi-insulating SiC need to be precisely controlled. Typically, n-type SiC is obtained by doping with N or P [131], p-type SiC by doping with Al or B [132,133], and semi-insulating SiC by doping with V [134–136].

Liu et al. [137] conducted nanoindentation tests on 4H-SiC with a peak load (P_{max}) ranging from 50 to 500 mN. Hardness and elastic modulus of 4H-SiC decrease with increasing indentation load, and that the hardness and elastic modulus of 4H-SiC decrease in the order of V-doped, undoped, and N-doped, as shown in Fig. 22(a) and (b). This explains why N-doped 4H-SiC microspheres are easier to slice to grind away the damage caused by slicing [138]. The decrease in modulus with increasing indentation load is ascribed to the combined effect of surface roughness, tip roundness and tip area function [139].

Sugiura et al. [140] established a piezoresistive mobility model and derived the relationship between ionization energy (E_{AI}) and doping concentration as:

$$E_{AI} = \frac{265}{1 + (N_A/1 \times 10^{19})} \quad (24)$$

Where, N_A is the doping concentration.

The results show (as shown in Fig. 23) that the ionization energy data are concentrated in the middle-heavy doped region. It can be seen that the change in ionization energy begins at about $1 \times 10^{18} \text{ cm}^{-3}$, and the proposed ionization energy model is considered to be sufficiently reliable [141]. The value of R^2 obtained is 0.929, which proves the validity of the proposed expression. This model can be used to determine the variation of ionization energy with Al doping concentration, but further researches are needed to predict the ionization energy of acceptor doping in WBG semiconductors at low temperatures.

A specific doping concentration can effectively improve the quality and reliability of SiC devices, but understanding how much impurity is

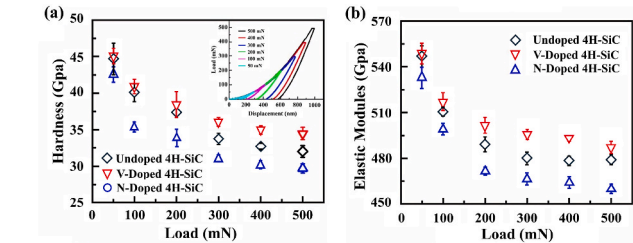


Fig. 22. Average values of (a) hardness and (b) elastic modulus with indentation depth with undoped, V-doped and N-doped 4H-SiC [137].

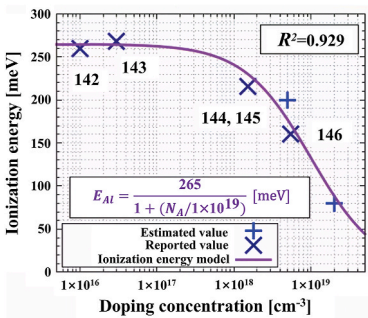


Fig. 23. Variation of ionization energy with doping concentration for Aluminum doping in 3C-SiC. Reported values (after Ref. [142–146]) are also shown [140].

doped into SiC is a prerequisite for obtaining a specific doping concentration. It is also important to test the quality of the doped SiC. Therefore, in order to obtain qualified SiC wafers, precise and practical testing technology is necessary, which can provide advantages for mass production of high-quality SiC devices [147]. Fig. 24 shows the commonly used characterization methods for single crystal SiC doping.

5. Preparation of SiC from silicon cutting waste

5.1. Silicon cutting waste generation and composition

With the rapid development of the photovoltaic industry, the demand for silicon wafers, which are indispensable for the manufacture of solar cells, has been increased dramatically. The production of silicon wafers requires the ingot to be first cut into square rods, and then the rods are sliced into 0.18–0.19 mm polycrystalline silicon wafers or monocrystalline silicon wafers [148]. There are two main types of crystalline silicon cutting technology: Slurry Sawing (SS) and Diamond Wire Sawing (DWS), as shown in Fig. 25 [149]. The principle of the SS cutting method is as follows: a water-based cutting fluid is composed of SiC particles as abrasives, polyethylene glycol (PEG) is adopted as a dispersant, and water as a solvent. The SiC particles are driven by a metal wire to roll and grind between the metal wire and the silicon ingot to cut the SiC crystals [150]. The principle of the DWS cutting method is that the diamond is attached to the metal wire by electroplating or resin bonding to form a diamond sawing wire. The cutting fluid is a mixture of water and a certain amount of organic solvent. The diamond wire is directly applied to the silicon ingot [151]. There are two types of silicon cutting waste (SCW) produced by the two different cutting methods: slurry sawing silicon waste (SSSW) and diamond wire sawing silicon waste (DWSSW). The main components of SSSW are a polyethylene glycol solution, metal impurities and almost equal amounts of Si powder and SiC particles [152]. DWSSW consists of a water-based coolant, a high concentration of silicon powder (more than 90%, without water)

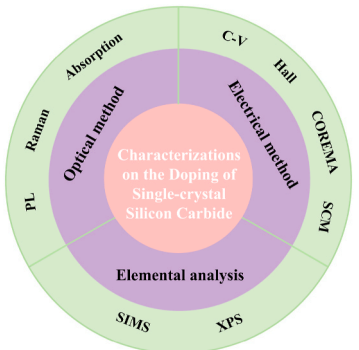


Fig. 24. Common characterization methods for SiC doping [147].

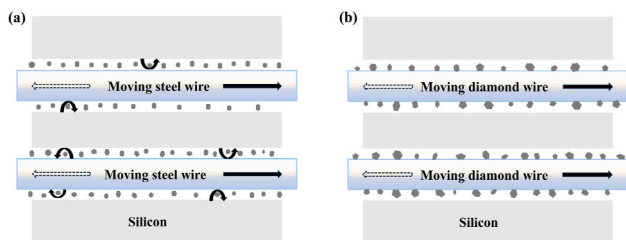


Fig. 25. Silicon ingot sawing techniques: (a) slurry sawing (SS); (b) diamond wire sawing (DWS) [149].

[153], a low level of metal impurities, a small amount of diamond particles and SiC. Since the cutting process is sufficiently cooled, the components in the slurry do not diffuse at high temperatures [154]. Due to the relatively complex composition of SSSW, there have been less researches on recycling methods for SSSW than for DWSSW. DWS is considered to be a significantly better alternative to SS [153,155]. At present, DWS has basically replaced SS for cutting silicon ingots.

The characterization of SSSW is shown in Fig. 26. The particle size analysis in Fig. 26(a) shows that SSSW is a fine particle, concentrated mainly around 2 mm, and the particle size distribution is mainly concentrated at 1–100 mm. The XRD analysis in Fig. 26(b) shows that the main components of the SSSW contain impurities of SiC, Si and Fe₂O₃. The FTIR analysis shown in Fig. 26(c) confirms the presence of SiO₂ and organic solvents introduced during the cutting process. Fig. 26(d) shows the chemical composition analysis of SSSW, which shows that SSSW is mainly composed of 21.3% SiC, 21.3% Si, 33.8% SiO₂, 10.1% metal oxide (Fe_xO_y), 10.9% PEG and 2.6% moisture [156].

The results of the characterization of the DWSSW are shown in Fig. 27. As shown in Fig. 27(a), the average particle size of the DWSSW powder can be observed to be approximately 1.344 μm. The XRD pattern shown in Fig. 27(b) indicates that the main phase in DWSSW is Si. The SEM image of the DWSSW in Fig. 27(c) demonstrates that the particles are present in the form of slices and agglomerates. The EDS spectrum in Fig. 27(d) shows that the main element in the DWSSW is Si, and the other main elements are O and Al. O is introduced by oxidation during the cutting process [157].

5.2. Preparation of SiC from SSSW

Although the cutting process has been sufficiently cooled, the purity of the Si powder is not affected by the high temperature diffusion of the components in the waste slurry. However, it is very difficult to separate Si powder with a size of less than 5 μm from SiC particles of similar size [158]. At the same time, due to the similarity of the physical and

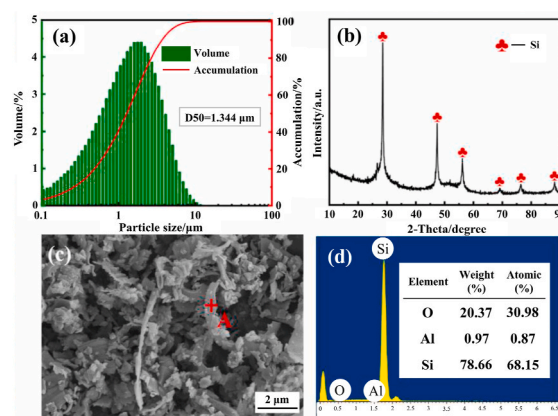


Fig. 27. Characterization of DWSSW: (a) particle size distribution; (b) XRD analysis; (c) SEM analysis; (d) EDS analysis [157].

chemical properties of Si and SiC, as well as the complex composition of the waste, it is difficult to separate the two types of micro-powder [159]. Therefore, some researchers have proposed to remove PEG and metal impurities from cutting waste, and instead of further separating Si and SiC, SiC is directly processed and prepared from the mixed waste of Si and SiC as the main raw material.

Cao et al. [160] used ethanol (C₂H₆O) and hydrochloric acid (HCl) to pretreat the Si powder in the silica slurry waste as the Si source and used coconut shell fiber as the biological C template. With the adoption of the spark plasma assisted chemical vapor deposition method, SiC fibers were prepared at 1000–1650 °C under an Ar atmosphere at a heating rate of 100 °C/min and a holding time of 10 min. The minimum reflection loss (RL) of the prepared SiC fibers was −21.88 dB, and the bandwidth of RL < −5 dB was about 1 GHz. Compared with the traditional CVD process, the method used in this study is a low-cost, time-saving, and simple technique. Gao et al. [156] used hydrochloric acid-treated silicon cutting waste as the Si source, petroleum coke (average particle size about 10 mm) as the C source, and a graphite crucible (purity 99.99 wt %) as the reaction vessel. The material was heated and cooled in an induction furnace, then annealed in a muffle furnace for 2 h to remove excess carbon, and finally acid-leached to obtain high-purity SiC (> 93 wt%). Fe₂O₃ was found to act as a catalyst to speed up the reaction and reduce the reaction temperature, but excess Fe can hinder the improvement of SiC purity. NaCl helps impurities to convert to volatile FeCl₃ and FeCl₂, and excess C can effectively promote the formation of SiC. Ren et al. [161] reported an environmentally friendly and efficient technology that uses silicon cutting waste as a Si source after dilution and acid washing, sisal fiber as a C source, and a closed graphite crucible containing sisal fiber and raw material powder. The crucible is heated to 800–1650 °C in an Ar atmosphere using a spark plasma system, and finally ultra-fine SiC crystals of about 200 nm are prepared by a spark plasma-assisted SiC reaction. The PL spectra of the ultrafine SiC crystals prepared show great application value in the field of optical devices such as blue-violet emitting devices. Jiang et al. [159] used silicon cutting waste as the Si source and petroleum coke (purity > 90.00%, D50 = 1.03 μm) as the C source. After planetary ball milling for 6 h, the mixed powder was granulated with a binder and melted in a high-purity graphite crucible (99.99%) in an atmospheric medium-frequency induction furnace (1600–2000 °C). The resulting product was subjected to milling, decarburization (95.20% removal rate) and ultrasonically assisted leaching to remove iron (99.38% removal rate), and dehydration to obtain high-purity SiC (> 98.9%). This process could effectively recover silicon from waste and optimizes leaching conditions. Ultrasonically assisted leaching kinetic analysis shows that the leaching process conforms to the chemical reaction control model. Fig. 28 shows the process flow diagram.

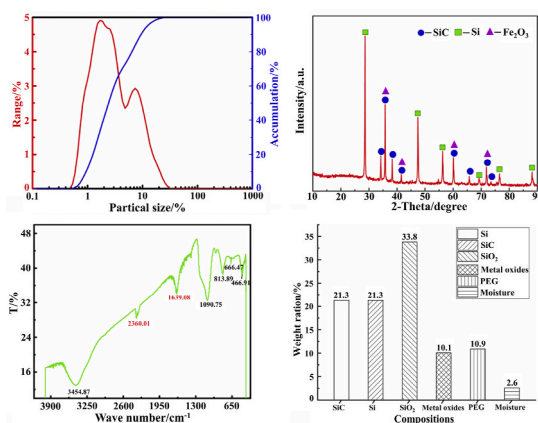


Fig. 26. Characterization of SSSW: (a) particle size distribution; (b) XRD analysis; (c) FTIR analysis; (d) chemical analysis [156].

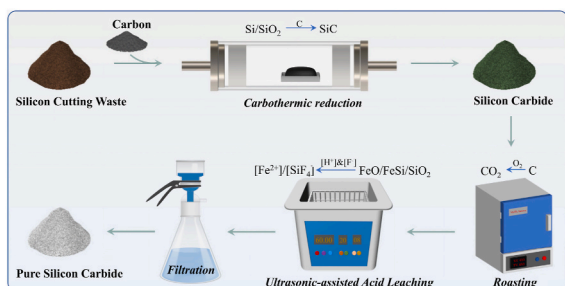


Fig. 28. Process flow diagram for the preparation of pure SiC from silicon cutting scrap [159].

5.3. Preparation of SiC from DWSSW

DWS is endowed with the advantages of fast cutting speed, low loss of single pieces, and more environmentally friendly cutting fluid, and is gradually being promoted and used by companies all over the world. Compared with SSSW, DWSSW is more inclined to separation and purification. At present, most researches are focused on purification methods to obtain high-purity silicon, including laser-assisted vacuum smelting [162], two-step sintering and acid leaching [163], alkaline dissolution [164], directional solidification [165,166], and other processes, while relatively few researches have been conducted on other potential applications. This summary will introduce the main researches on the preparation of SiC with the adoption of DWSSW.

Hossain et al. [167] adopted DWSSW as the Si source and high-purity carbon powder as the C source (99.99%, < 325 mesh). The DWSSW and C powder were mixed by ball milling and then formed particles by a grinding tool and a press. The particles were then placed in an alumina crucible and transferred to a reaction tube with Ar flowing through it for reaction. Finally, the product was cooled to room temperature and ground into powders. The experimental setup is shown in Fig. 29. The formation of SiC was confirmed by XRD, SEM and EDS analysis, and ATR analysis of the samples after heat treatment confirmed the removal of C from the SiC powder.

Jiang et al. [168] employed silicon cutting waste and sucrose as raw materials, dissolved the raw materials in deionized water to form a suspension, then heated it to obtain a dry black solid, placed it in an alumina crucible, which was subsequently heated in a low Ar atmosphere tube furnace, and later subjected to cooling, pickling, purification and drying processes. Spherical SiC nanoparticles with a particle size of 40–50 nm were successfully prepared, which exhibit special photoluminescence properties and good thermal stability. A comparison of the composition analysis of silicon cutting waste was conducted by Jiang et al., and the characterization results in Fig. 27 show that the silicon cutting waste used by their team is DWSSW. In 2023, the Jiang's [169] team proposed a new method for the simple and sustainable preparation of nano-SiC using DWSSW. By adopting DWSSW as the silicon source and phenolic resin (analytical grade, > 99 wt%) as the

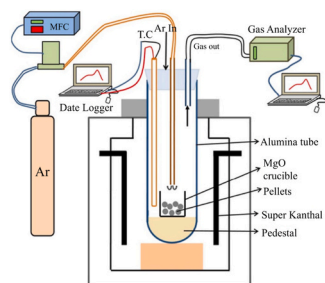


Fig. 29. Schematic diagram of the experimental decorations used by Hossain et al. [167].

carbon source, the raw materials were mixed uniformly and placed in an alumina crucible. The mixture was then heated and melted in a horizontal tube furnace with Ar flow, and the product was subsequently acid-leached to obtain nano-sized homogeneous SiC particles (average diameter about 30 nm) with good oxidation resistance. The formation of nano-sized homogeneous SiC particles could be ascribed to the dispersion of phenolic resin during the initial mixing process, the micron-sized particles of DWSSW, and the pyrolysis of phenolic resin. Fig. 30 shows a schematic diagram of the reaction mechanism of silicon cutting waste and phenolic resin to synthesize nano-sized SiC.

The main advantage of adopting SSSW as the main raw material to produce high-purity SiC and ultrafine SiC is that the step of separating Si and SiC crystals is eliminated, thereby reducing the cost of recycling, and to a certain extent avoiding environmental pollution caused by waste. By adjusting the proportion of raw materials, the reaction of the Si powder in the cutting waste with C is ensured to be completely converted into SiC, thereby increasing the recovery efficiency of Si crystals and the output of SiC crystals. The prepared SiC products have high utilization value. Compared to the traditional industrial SiC preparation method, this method not only has lower raw material costs and a lower reaction temperature, but also simplifies the entire process and enables the comprehensive reuse of SSSW.

The combined physical and chemical method of preparing and purifying SiC opens up new paths for the recycling of DWSSW, and could maximize the economic added value of cutting waste. The preparation of SiC from DWSSW makes full use of the high content of Si in the waste, and the addition of a sufficient amount of C source can produce a large amount of SiC. At the same time, since the impurity content in DWSSW is relatively low, and the impurity composition is relatively simple, the treatment of impurities can be reduced, which could effectively simplify the entire reaction process. However, there still exist some shortcomings such as high smelting temperature, high energy consumption and low purity of purification.

At present, the SiC prepared by SCW exhibits excellent properties such as small particle size, large bandwidth, non-toxicity, good biocompatibility and antioxidant properties, etc [169]. However, the low purity and other problems limit its application, and it is only used in ceramic materials at present [170,171]. Although the use of SCW to prepare SiC plays a great role in resource recycling and reuse, this study is still limited to dealing with the research stage, and there is no report on the market, cost and predictable profit of the recycling technology of SCW prepared SiC.

6. Conclusions

This paper reviews the research progress of SiC, including the crystal structure of SiC, the growth method of SiC crystals, the unique properties, and the preparation of SiC materials from silicon cutting waste. The

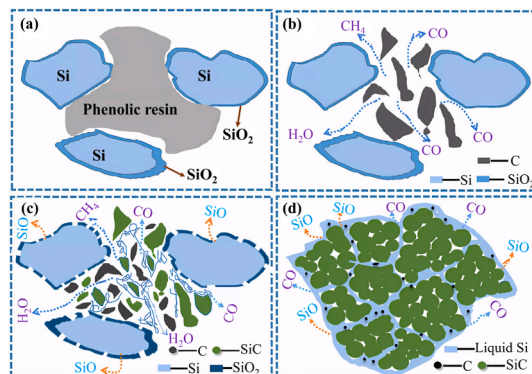


Fig. 30. Schematic diagram of reaction mechanism of synthesis of nano-structured SiC using silicon cutting waste (SCW) and phenolic resin [169].

main methods currently adopted for SiC crystals growth include PVT, HTCVD and TSSG. TSSG is capable of growing low defects, tubeless SiC crystals at lower temperatures, and is one of the new technologies for growing low-cost, high-quality SiC crystals. The stable crystal structure of SiC makes it an excellent material with great application value in the high-temperature and high-frequency field. However, since SiC has a variety of crystal structures, its performance may vary greatly. It is necessary to establish a complete system of SiC polycrystalline structure performance to make it perform better in various applications. For economic and environmental reasons, there is a growing demand to recycle the rapidly increasing amount of silicon waste from the photovoltaic industry. At present, the purpose of most researches on SCW is mainly focused on separation and purification, and there are few researches on the preparation of SiC from SCW. Compared with the traditional industrial method of preparing SiC from silicon cutting waste, the cost of preparing SiC raw materials is lower, the processing of impurities is reduced, and the reaction temperature is also lower. However, the recovery and preparation processes need to be optimized so as to ensure the quality and recovery rate of the product.

Declaration of Competing interest

Approved by all authors, no conflict of interest exists in the submission of this manuscript. On behalf of all co-authors, I would like to declare that this work was the original research that has not been published or under consideration for publication elsewhere.

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